

AUTONOMIC PHARMACOLOGY

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2.1 NEURO-HUMORAL TRANSMISSION IN THE AUTONOMIC NERVOUS SYSTEM

NEUROTRANSMISSION IN THE AUTONOMIC NERVOUS SYSTEM

INTRODUCTION:

- The principal neurotransmitters in the ANS are acetylcholine and noradrenaline
- Acetylcholine is the neurotransmitter at all autonomic ganglia, neuromuscular junctions and parasympathetic tissue synapses
- All preganglionic neurons and parasympathetic postganglionic neurons use acetylcholine as neurotransmitter

NEUROTRANSMISSION IN THE AUTONOMIC NERVOUS SYSTEM

INTRODUCTION ... CONT'D:

Most postganglionic sympathetic neurons use norepinephrine (noradrenaline) as the neurotransmitter, EXCEPT:

- Sweat glands: Innervation of the thermoregulatory sweat glands is anatomically sympathetic, but the postganglionic neurons release acetylcholine as the neurotransmitter
- Kidneys: Postganglionic neurons to the smooth muscle of the renal vasculature release dopamine

NEUROTRANSMISSION IN THE AUTONOMIC NERVOUS SYSTEM

INTRODUCTION CONT'D:

- Adrenal gland: Preganglionic neurons synapse directly on the adrenal gland, release acetylcholine, and activate nicotinic acetylcholine receptors in the adrenal gland medulla. The adrenal glands release epinephrine (adrenaline) into systemic circulation

NEUROTRANSMISSION IN THE AUTONOMIC NERVOUS SYSTEM

LS - 16: LEARNING OBJECTIVES:

- To describe the neurotransmitter systems in the autonomic nervous system
- To outline the synthesis, storage, release, metabolism and uptake of the principal neurotransmitters in the autonomic nervous system

KEY DEFINITIONS

Neurotransmitter: A chemical that transmits signals from one neuron to another or from a neuron to an effector cell

Synapse: A junctional connection between two neurons, across which a signal can pass

Pre-synaptic neuron: Where a neurotransmitter is synthesized, stored and released upon cell activation

Post-synaptic neuron or effector cell: Where neurotransmitter is detected and its action translated into cellular activities

NEUROTRANSMISSION IN THE PARASYMPATHETIC NERVOUS SYSTEM (PNS)

- The principal neurotransmitter in the parasympathetic nervous system is acetylcholine
- There are two types of acetylcholine receptors: nicotinic (N) and muscarinic (M)
- Neurotransmission in the ganglia is achieved by preganglionic fibers releasing acetylcholine into the synaptic space, which activates nicotinic receptors on postganglionic nerve dendrites

NEUROTRANSMISSION IN THE PNS ... CONT'D

- Depolarization of postganglionic cholinergic nerves results in the release of their stored acetylcholine within the innervated tissue. The acetylcholine activates muscarinic receptors on effector cells to produce biological effects
- There are 5 sub-types of muscarinic receptors: M_1 , M_2 , M_3 , M_4 and M_5
- The release of acetylcholine is regulated by acetylcholine acting on presynaptic M_2 receptors – activation of these receptors inhibits release of acetylcholine

ACETYLCHOLINE (Ach)

- Ach is synthesized in nerve terminals by the cytoplasmic enzyme choline acetyltransferase which catalyses the transfer of an acetate group from acetyl coenzyme A to choline that has been transported into cholinergic neurons by a sodium dependent membrane carrier
- The synthesized Ach is transported from the cytoplasm to storage vesicles
- Ach is released from the vesicles by nerve action potentials through calcium-dependent exocytosis. Release can be blocked by botulinum toxin.

ACETYLCHOLINE CONT'D

- The effect of ACh on postsynaptic tissue is terminated by rapid action of acetylcholinesterase (AChE) that metabolizes ACh to acetate and choline. The liberated choline is taken into the presynaptic nerve terminal.
- ACh is not administered for therapeutic purposes because it is hydrolysed almost instantly by butyrylcholinesterase in plasma (pseudocholinesterase, plasma cholinesterase)

NEUROTRANSMISSION IN THE SYMPATHETIC NERVOUS SYSTEM (SNS)

- Preganglionic neurons (synapse with postganglionic neurons in the ganglia) release acetylcholine as the primary neurotransmitter
- Postganglionic neurons release norepinephrine as the primary neurotransmitter

NEUROTRANSMISSION IN THE SNS CONT'D

Synthesis of catecholamines (dopamine, norepinephrine and epinephrine)

1. In presynaptic nerves, tyrosine is hydroxylated by tyrosine hydroxylase to form dihydroxyphenylalanine (dopa)
2. Dopa is then decarboxylated by dopa decarboxylase to form dopamine
3. Dopamine is then transported into vesicles (this step is blocked by the drug reserpine), where it is hydroxylated to form norepinephrine

In certain areas of the brain and in the adrenal medulla, norepinephrine is methylated to epinephrine

NOREPINEPHRINE STORAGE AND RELEASE

- Norepinephrine is stored in the vesicle with ATP and neuropeptide Y (co-transmitters, neuromodulators)
- Norepinephrine is released into the synaptic cleft by exocytosis (a calcium-dependent process) when the nerve terminal is depolarized

NOREPINEPHRINE CONT'D

The released norepinephrine interacts with specific adrenergic receptors (adrenoceptors) to produce tissue/organ response:

- Postsynaptic response
- Presynaptic modulation of further neurotransmitter release

Norepinephrine release can be blocked by such drugs as bretylium and guanethidine

NOREPINEPHRINE AND EPINEPHRINE RELEASE

- Norepinephrine also exists in a non-vesicular cytoplasmic pool that is released by indirectly acting sympathomimetic amines (e.g. tyramine, amphetamine, ephedrine) by a process that is not calcium-dependent.
- Norepinephrine and some epinephrine are released from adrenergic nerve endings in the brain
- In the periphery, epinephrine is the major catecholamine released from the adrenal medulla into the general circulation, where it functions as a hormone

TERMINATION OF NOREPINEPHRINE ACTION

- The action of norepinephrine is terminated primarily by active transport into the cytoplasm of the pre-synaptic neurone (uptake-1)
 - This process can be inhibited by cocaine and tricyclic antidepressants such as amitriptyline
- Norepinephrine is then transported by a second carrier system into storage vesicles
 - This process can be blocked by reserpine

TERMINATION OF NOREPINEPHRINE ACTION CONT'D

- Norepinephrine is also taken into post-synaptic tissue (uptake-2)
- In the cytoplasm of pre-synaptic nerve cells, synaptic cleft and in post-synaptic tissues, norepinephrine is metabolized by monoamine oxidase (MAO) or catechol-O-methyltransferase (COMT) to inactive intermediates and finally to homovanillic acid (HVA) and methoxyhydroxy mandelic acid (VMA)

REGULATION OF NORADRENALINE RELEASE

Pre-synaptic receptors on sympathetic nerve terminals influence the release of norepinephrine:

- α_2 -adrenergic and muscarinic M_1/M_2 receptors decrease norepinephrine release
- AT_1 -angiotensin and nicotinic receptors increase norepinephrine release

LEARNING OUTCOMES

Students must be able to comprehend:

- The neurotransmitter systems in the autonomic nervous system
- The synthesis, storage, release, metabolism and uptake of the principal neurotransmitters in the autonomic nervous system

END

Thanks for listening